Neutron Transport and Tritium Breeding Modelling in Nuclear Fusion Reactor Breeder Blankets

MPhys Project Report

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Abstract

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CONTENTS

1. INTRODUCTION

Start your new document here! Moir, 1982, Momoshima (2022), (J. D. Lee, 1983), (Segantin et al., 2020)

2. THEORY

2.1. Nuclear Physics

2.1.1. Relevant Nuclear Reactions

While lead and beryllium are deliberately selected as primary multipliers due to their high $(\mathrm{n}, 2\,\mathrm{n})$ cross-sections (Hernández and Pereslavtsev, 2018), structural materials within the blanket also exhibit minor multiplication properties at fusion energy scales. For instance, the dominant isotope in structural steels like EUROFER97 (Stornelli et al., 2024) is $^{56}\mathrm{Fe}$, which possesses a small but non-zero $(\mathrm{n}, 2\,\mathrm{n})$ cross-section for 14.1 MeV neutrons (Brown et al., 2018). This threshold reaction proceeds as:

$$^{56}\mathrm{Fe} + \mathrm{n} \longrightarrow {}^{55}\mathrm{Fe} + 2\,\mathrm{n} \tag{1}$$

2.1.2. Neutron Track Length

To quantify the neutron population within specific geometric regions of the blanket, stochastic Monte Carlo transport codes, such as OpenMC (Romano et al., 2015), utilise a track-length estimator. Rather than counting individual neutrons crossing a boundary, the simulation integrates the total distance traversed by all particle histories within a defined cell volume (Kalos and Whitlock, 2008).

This yields the volume-integrated scalar flux (Φ_V), which is expressed in units of neutron-centimetres ($\mathrm{n} \cdot \mathrm{cm}$) and is formally defined as (Kuridan, 2023):

$$\Phi_V = \int_V \int_E \int_{4\pi} \psi(\vec{r}, \hat{\Omega}, E)\, d\hat{\Omega}\, dE\, dV = \sum_i W_i \ell_i \tag{2}$$

where W_i is the statistical weight of the neutron and ℓ_i is the path length of its trajectory through the volume V.

A longer total neutron path length implies a longer residence time within the breeder material, which directly increases the probability of nuclear interactions. The macroscopic reaction rate for any process, such as elastic scattering, is simply the product of the region's macroscopic cross-section (Σ_x) and this volume-integrated flux (Kuridan, 2023).

2.1.3. Radiation Damage and Displacements Per Atom

To accurately assess the structural viability of the breeder blanket and first wall, the neutronic simulation tallies radiation-induced material degradation. The standard metric for quantifying this degradation in a crystalline lattice is Displacements Per Atom (DPA), which represents the average number of times an atom is knocked out of its equilibrium lattice site due to radiation bombardment.

When a high-energy fusion neutron collides with a target nucleus, it transfers kinetic energy, creating a Primary Knock-on Atom (PKA). If the transferred energy exceeds the lattice displacement threshold (E_d), the PKA is ejected and can trigger a secondary collision cascade. However, not all the kinetic energy transferred to the PKA contributes to atomic displacements; a significant fraction is lost to inelastic electronic excitation (ionisation) as the PKA travels through the lattice. The specific fraction of the transferred energy available to cause further elastic nuclear collisions and subsequent atomic displacements is tallied as the "Damage Energy" (T_d).

In the standard Norgett-Robinson-Torrens (NRT) displacement model (Norgett, Robinson, and Torrens, 1975), the raw number of displaced atoms (N_d) is directly proportional to this damage energy:

$$N_{d,\mathrm{NRT}} = \frac{0.8T_d}{2E_d} \tag{3}$$

While the NRT model provides a reliable baseline, it often overestimates the final number of stable defects because it does not account for the athermal recombination of atoms that quickly regain their original lattice positions. To correct for this phenomenon, the Athermic Recombination-Corrected (arc-dpa) model is employed (Nordlund et al., 2018). This introduces a surviving defect fraction, $\xi(T_d)$:

$$N_{d,\mathrm{arc}} = \frac{0.8T_d}{2E_d} \cdot \xi(T_d) = \frac{0.8T_d}{2E_d}\left((1-c)\left(\frac{0.8T_d}{2E_d}\right)^b + c\right) \tag{4}$$

where b and c are material-specific constants dependent on the displaced atom type. Values for the displacement energy (E_d) and these constants for critical structural materials, such as iron (dominating the structural separators) and zirconium (in the first wall), are established in the literature (Saha, Devan, and Ganesan, 2019).

Finally, to determine if a material exceeds its irradiation limits over its operational lifetime, the total number of displacements must be normalised into DPA per Full Power Year (FPY). The DPA received by a structural component over a full-power year is given by:

$$\mathrm{DPA/FPY} = \frac{N_{d,\mathrm{arc}}}{\left(\frac{\rho V \cdot N_A}{\mu}\right)} \cdot \frac{P}{E_f \cdot e} \cdot (60^2 \cdot 24 \cdot 365) \tag{5}$$

Here, the denominator calculates the total number of target atoms within the specific component using the material density (ρ), volume (V), Avogadro's constant (N_A), and molar mass (μ). This is multiplied by the annual neutron emission rate—derived from the reactor thermal power (P) and the energy released per fusion event (E_f, converted to Joules via the elementary charge e)—scaled to the number of seconds in a calendar year. Tracking this rigorous DPA metric allows for an accurate evaluation of the severity of the neutron flux on the structural boundaries.

2.1.4. Nuclear Heating and Energy Deposition

In addition to radiation damage, the structural and thermal-hydraulic viability of a blanket design is dictated by the spatial distribution of nuclear heating. Energy is deposited into the blanket materials through two primary mechanisms: neutron kinetic energy transfer during elastic and inelastic scattering, and the absorption of secondary gamma rays (γ-photons) produced during nuclear reactions (Kuridan, 2023).

In neutronic simulations, this localised energy deposition is calculated using Kerma (Kinetic Energy Released in Materials) factors. The total nuclear heating rate, $H(\vec{r})$, at a specific spatial location is the integral of the local scalar flux $\Phi(\vec{r}, E)$ folded with the macroscopic Kerma factor $K(E)$ over all energies (Kuridan, 2023):

$$H(\vec{r}) = \int_E \Phi(\vec{r}, E) K(E)\, dE \tag{6}$$

where $K(E)$ accounts for the sum of all microscopic energy-releasing reaction cross-sections multiplied by the average energy released per reaction. Accurately mapping this heating distribution is critical, as it determines the required mass flow rates and spatial distribution of the coolant layers to prevent material melting and ensure efficient thermal energy extraction for the power cycle. Advanced Monte Carlo codes, such as OpenMC, directly implement these integrals via specialised `heating` tallies to provide high-fidelity energy deposition profiles (Romano et al., 2015).

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4. METHODOLOGY

4.1. Simulation Pipeline Summary

The neutronic performance of the breeder blanket geometries is evaluated using a coupled computational workflow. The 3D parametric CAD models of the tokamak reactor are generated using the Python-based framework Paramak (Shimwell et al., 2021). To ensure computational efficiency while maintaining physical accuracy, the simulation domain comprises a 90° toroidal sector with periodic boundary conditions applied to the bounding surfaces. These Constructive Solid Geometry (CSG) geometries are exported as Direct Accelerated Geometry Monte Carlo (DAGMC) unstructured meshes, which allow for efficient ray-tracing directly on the CAD surface representations without the need for voxelisation. Stochastic neutron transport simulations are subsequently executed upon these meshes using OpenMC (Romano et al., 2015). To ensure statistical robustness and minimise relative error below 1%, simulations utilised high particle counts ranging between 10^5 and 10^7 neutrons per run.

Based on the optimal configurations identified in the previous semester, the material definitions for this architecture are explicitly fixed: the first wall utilises a zirconium alloy, the breeder is a lithium-lead eutectic ($Li_{17}Pb_{83}$), and the coolant is pressurised water (modelled at 600 K and 15.5 MPa). EUROFER97 was selected for the internal structural separators, rear wall, and central solenoid, reflecting industry standards for robust, reduced-activation ferritic-martensitic steels. Furthermore, to more accurately evaluate the spatial distribution of interactions within this refined geometry, the 14.1 MeV Muir-distributed ring source utilised in the previous study has been updated to a uniform volumetric body source.

A comprehensive exposition of this foundational workflow was established in the first stage of this study (Khan, 2026a). Building upon those established homogeneous baseline models, this current study transitions to a discrete, heterogeneous layered blanket architecture. We use 4 breeder layers and 4 coolant layers, with separators between each breeder-coolant interface (see Figure 1). This addresses the core unrealistic homogenous approximation of that study, establishing a key step towards more realistic breeder blanket module requirements and challenges. Note that other studies have simulated detailed models with specific coolant piping interlaced in the blanket (see Section 3.2). Paramak's parameterised framework limits the level of heterogeneity to systems with azimuthal symmetry (such as our layered model). The layered model is still an approximation compared to specific real implementations. We proceed as there are key advantages to this approach:

- **Conservative heterogeneity:** Applying this model allows for the majority of heterogeneous effects to be modelled. Importantly, in real implementations streaming effects

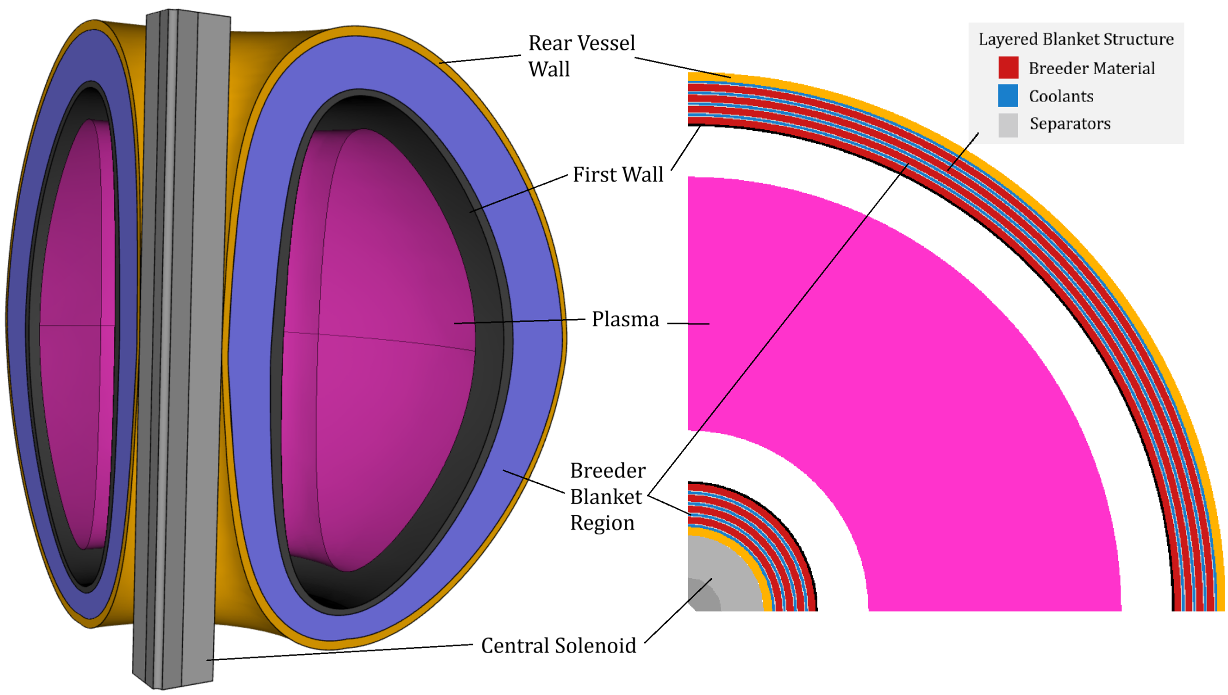


Figure 1: Representative geometry of the simplified Tokamak model generated via Paramak. The simulation domain comprises a 90° revolution with periodic boundary conditions. Included is an example breakdown of the layered blanket, only the coolant and breeder thicknesses are adjusted in this study.

allow some neutrons to bypass the interactions between coolants and separators. The layered model prevents this, forcing heterogenous material paths. Given heterogeneity negatively affects performance (see Figure 5a), the layered model notably becomes a conservative estimate of blanket performance. In an immature field where specific designs are undecided, a conservative study can be informative.

- **Parametrisation:** A layered model is easily parameterised into a set of thickness variables for each layer. Comprehensively exploring and optimising even this 8-layer model required complex techniques that became a majority component of this project (see Section 4.3). Using a more detailed model necessitates a greater number of parameters and thus exploring higher dimensional spaces. This is non-trivial to optimise, it is instructional to begin with a lower dimensional parameterised model before scaling.

4.2. Neutronic Analysis

4.2.1. Mechanistic Analysis of the Layered Blanket

Initial parametric sweeps were conducted to establish a baseline for the heterogeneous architecture. We also built upon explorations in the first stage of this project (Khan, 2026a) by including tallying of individual layers to understand which were contributing and impactful to global results (see Section 5.1).

Building on this further, mechanistic understanding of performance necessitates localised analysis of several neutron transport parameters throughout the reactor. To generate two-dimensional spatial distributions of nuclear heating and radiation damage, regular Cartesian mesh tallies were overlaid onto the geometry of the blanket model.

In OpenMC, a high-resolution 2D bounding box mesh was defined across the poloidal-radial plane of the blanket segment. Within this mesh grid, six specific reaction metrics were scored independently:

- **Neutron Flux:** Scored using the `flux` tally, measuring the volume-integrated track length of all neutrons across the mesh to map the spatial attenuation of the neutron population.

- **Scattering Density:** Scored using the `scatter` tally, mapping the density of elastic and inelastic collision events primarily responsible for neutron moderation.

- **Tritium Production:** Scored using the `H3-production` tally, identifying the specific geometric regions within the breeder blanket that contribute most heavily to the global TBR.

- **Neutron Multiplication:** Scored using the `multiply` tally, highlighting the precise locations of high-energy threshold reactions (such as $(\mathrm{n}, 2\,\mathrm{n})$ in the multipliers and structural separators) that boost the neutron economy.

- **Nuclear Heating:** Scored using the standard `heating` tally, which records the total kinetic energy deposited by both neutrons and secondary photons per source particle.

- **Damage Energy:** Scored using the `damage-energy` tally, capturing the specific fraction of kinetic energy available to cause atomic displacements in the structural lattice.

The raw outputs from these mesh tallies were subsequently processed and visualised as heat maps (see Section 5.1).

4.2.2. Manual Thickness Distribution Analysis

To systematically investigate the effect of radial material distribution within the heterogeneous blanket, a constrained parametric approach was initially required. The full layered architecture contains eight independent geometric variables (comprising four discrete breeder layers and four coolant layers). Exhaustively exploring this 8-dimensional design space via traditional manual or grid-search methods is computationally intractable.

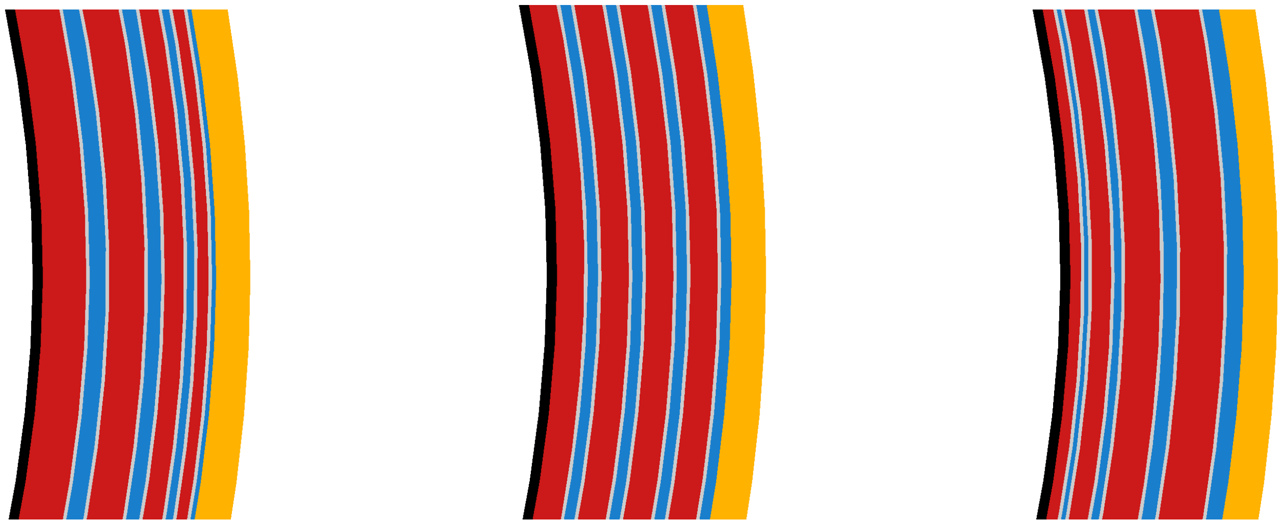
To reduce the dimensionality of the problem while maintaining physical relevance, a linear gradient parameter, ζ, was introduced as a heuristic constraint. This parameter governs the relative thickness of the layers across the blanket's depth, providing a human-intuitive method to observe spatial trends.

While this constrained, single-variable approach yields valuable insights into the fundamental physics of the blanket, it inherently restricts the exploration of the broader design space. The limitations of manual parametrisation in identifying true global optima across an unconstrained 8-dimensional parameter space necessitate the use of advanced computational techniques. Consequently, the study transitions to the application of Machine Learning algorithms (see Section 4.2) to navigate this complex parameter space efficiently.

4.3. ML Optimisation

4.3.1. ML Algorithm

To navigate the computationally expensive 8-dimensional design space effectively, Single-Objective Bayesian Optimisation (BO) was employed. The optimisation pipeline was developed using the Ax framework, leveraging BoTorch and PyTorch as the underlying computational engines. A Gaussian Process Regressor (GPR) was utilised as the surrogate model. This scalable, N-dimensional architecture is highly sample-efficient, capable of converging upon optimal configurations within $N < 150$ trials (tested in spaces up to 8 dimensions). This sample efficiency makes it ideal for optimising computationally intensive physical simulations. Furthermore, the objective function and hyperparameters are designed to be modularly configurable, with automated generation of surrogate parameter surfaces and optimisation progress diagnostics.



(a) Forward-weighted ($\zeta < 0$)

(b) Uniform/Baseline ($\zeta = 0$)

(c) Rear-weighted ($\zeta > 0$)

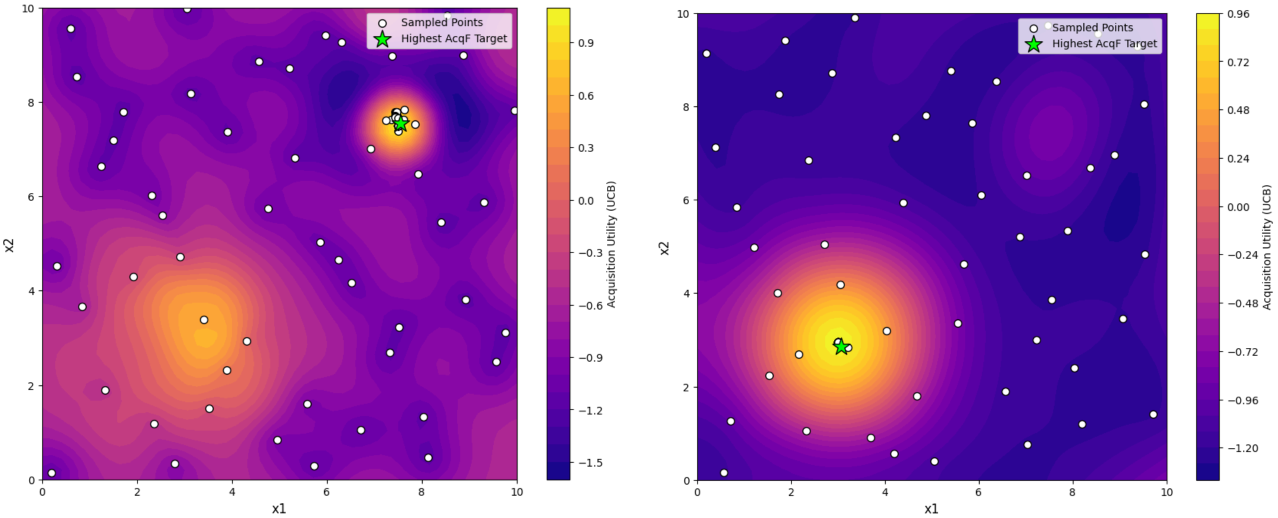
Figure 2: Cross-sectional visualisations of the breeder blanket layers illustrating the effect of the linear gradient parameter, ζ. This parameter controls the radial distribution of layer thicknesses: (a) a forward-weighted geometry shifting volume towards the first wall, (b) the baseline uniform thickness distribution, and (c) a rear-weighted geometry shifting volume deeper into the blanket.

The objective function was defined by wrapping the discrete parametric OpenMC neutronics simulation into an automated evaluation loop. This function takes the eight blanket layer thicknesses as continuous input parameters (bounded by defined physical engineering constraints) and returns the global TBR as the single scalar objective to be maximised. The computational overhead of the ML solver was negligible (≈ 20 s for $N = 150$ trials); thus, the total runtime was bounded entirely by the neutronics simulation, which required ≈ 60 s per trial on a standard desktop workstation. Because the optimiser was constrained to a maximum of $N = 150$ trials, all ML optimisation studies were completed in under 3 hours. The complete source code developed for this pipeline and complete optimisation results is publicly available (Khan, 2026b).

A critical component of this framework is the selection and tuning of the acquisition function, which dictates how the algorithm samples new data points. For this study, the Upper Confidence Bound (UCB) acquisition function was selected over Expected Improvement (EI). Unlike EI, UCB provides explicit, scalar control over the exploration-exploitation trade-off via the hyperparameter β. Properly tuning this parameter is vital for a high-dimensional physical system where multiple local maxima may exist. Figure 3 illustrates a 2D projection of the acquisition utility landscape used to benchmark the model's behaviour.

As demonstrated in Figure 3b, setting a low exploration weight ($\beta = 1$) results in a overly greedy search. The algorithm heavily exploits a known local optimum, entirely missing a secondary, hidden peak within the design space. Conversely, Figure 3a shows that a significantly higher parameter ($\beta = 1,000,000$) forces the model to prioritise high-uncertainty regions, successfully mapping the broader landscape and identifying the hidden peak.

Given the complexity and scale of the layered blanket parameter space, premature convergence on a local optimum poses a significant risk. Consequently, the UCB acquisition function was configured with a highly explorative $\beta = 1,000,000$ value to ensure robust, global mapping of the design space before narrowing in on the optimal layer thicknesses



(a) Explorative tuning ($\beta = 1,000,000$)

(b) Greedy tuning ($\beta = 1$)

Figure 3: Comparison of the Upper Confidence Bound (UCB) acquisition function behaviour under different exploration parameters (β). (a) A high β value successfully prioritises exploration, identifying a secondary hidden peak. (b) A low β value results in overly greedy exploitation, causing the model to stagnate on a single local optimum and ignore unexplored regions.

4.3.2. Testing, Tuning & Developing the Optimiser

Before applying the Bayesian optimiser to the high-dimensional neutronic blanket model it was evaluated against a suite of notoriously difficult mathematical landscapes, including the including the Rastrigin (Rastrigin, 1974), Schwefel (Schwefel, 1981), Rosenbrock (Rosenbrock, 1960), and Gramacy & Lee (Gramacy and H. K. Lee, 2012) functions, to rigorously benchmark the optimiser's robustness against stagnation. These functions were selected for their diverse pathological traits—such as high-frequency noise, deceptive local minima, and non-symmetric valleys. These functions were scaled incrementally from 1D up to 4D to ensure the ϵ-greedy and peak-hunting mechanisms remained effective as the search space expanded exponentially.

Figure 4a demonstrates a 1D test case. A critical distinction highlighted here is the difference between the "Actual Best Observed" (the highest discrete point successfully sampled during the simulation) and the "Best Model Prediction" (the theoretical peak calculated by the surrogate model). While finding the absolute optimal TBR is the primary engineering goal, the full continuous surrogate model is arguably the most valuable output for a physicist, providing physical intuition regarding how variables interact across the entire domain.

While 1D tests validate basic functionality, scaling the benchmark to 2D surfaces (Figure 4b, 4c, 4d) is necessary to ensure the algorithm does not fail as dimensionality increases. During these tests, it became apparent that standard Upper Confidence Bound (UCB) acquisition, even with a high exploration parameter, could occasionally stagnate in deep local minima. To prevent this, three bespoke algorithmic enhancements were introduced:

- **Interleaved random injection:** An ϵ-greedy approach was implemented, using a defined probability parameter to periodically force a completely random sample, ensuring exploration in areas the UCB function might otherwise ignore. As the GPR-BO is able to optimise local peaks quickly, we set found setting the probability to $\epsilon = 0.5$ optimal; particularly because we found that UCB acquisition "randomly explored" more in higher dimensions as the space gets exponentially larger.

- **Duplicate avoidance:** Once stuck in a local minimum, the raw BO algorithm would commonly repeatedly test the peak of that minima. Our optimiser was constrained to reject repeated coordinates, forcing continuous spatial exploration and preventing wasted computational resources on identical measurements.

- **Multimodal surrogate exploitation:** Every n_{hunt} trials, the algorithm actively scans the surrogate model to identify the top five predicted peaks. It then forces a sample at the highest peak that remains locally unexplored (highlighted by the pink markers in Figure 4d). In our trials we found $n_{hunt} = 10$ sufficient.

Successfully navigating these notoriously difficult mathematical landscapes demonstrates the optimiser's robust resistance to local minima. This is evidenced by its successful discovery of the 2D hidden peak (Figure 4c) and its highly accurate structural reconstruction of the complex, multi-modal 2D Schwefel function (Figure 4d).

The algorithm was potentially over-prepared for the physical breeder blanket investigation, which is expected to feature broader, continuous global physical trends rather than sharp, discontinuous mathematical spikes. As illustrated by the Gramacy & Lee benchmark (Figure 4b), the model efficiently isolates and solves for broad overall trends, naturally exhibiting less sensitivity to high-frequency, small-length-scale variations. This testing ultimately confirms that while the algorithm maps wider global dynamics highly effectively, microscopic local peaks represent the primary hazard where the model risks entrapment.

4.3.3. ML Optimisation Studies for TBR-Thickness

To formally bound the Bayesian optimiser against unrealistic values and meshing errors, dimensional limits were imposed on the blanket components based on engineering constraints. Discrete breeder layers were restricted to thicknesses between 3.0 cm and 16.0 cm, while coolant layers were bounded between 1.5 cm and 6.0 cm.

The model was applied on a parameterised version simulation code, in which it had control over the 8 breeder thicknesses (4 breeder layers, 4 coolant layers). It was given a single optimisation objective, to maximise TBR. Throughout testing, monotonic convergence was often observed (corner/boundary solutions experiencing binding constraints were common) when no additional constraints were applied. Therefore, in the final set of studies we present several optimisation runs in which we applied additional parameter constraints.

- **8D:** The model is free to adjust all 8 layers, as long as they are bounded within the provided domains.

- **8D Constrained:** A strict total blanket thickness budget of 44.0 cm.

- **4D Breeder Constrained:** Breeder layers were limited to a collective 32.0 cm budget, with all coolant layers locked at their 1.5 cm minimum limit.

- **4D Coolant Constrained:** Coolant layers were required to meet a minimum collective thickness of 12.0 cm, with breeder layers locked at 8.0 cm each.

To establish a useful comparison for the 44.0 cm budget, a manual "Binding" configuration was generated. Binding constraints / limits here were when we maximised the innermost plasma-facing breeder layers to their 16.0 cm upper bound while minimising the interleaving coolants to 1.5 cm. We would prioritise inner layers and continue radially outward until the minimum bounds were forced to meet the remaining thickness budget.

5. RESULTS

5.1. Layered Blanket Model Neutronics

Building upon the baseline parameter sweeps conducted in the first semester (Khan, 2026a), this phase of the project transitions the neutronic model from a simplified homogeneous approximation to a discrete, heterogeneous layered architecture.

Figure 5a presents the global TBR across these different modelling approaches. The data shows an initial reduction in the calculated TBR when the required structural separator material (Eurofer) is integrated into the homogeneous mixture. A further, more substantial decrease in the global TBR is observed when this exact same material volume is geometrically separated into distinct breeder, coolant, and structural layers.

To isolate the spatial distribution of tritium production within this heterogeneous geometry, the TBR was tallied layer-by-layer as a function of the total radial blanket thickness (Figure 5b). For this parameter sweep, the thickness of the structural separators was held constant while the radial depths of the discrete breeder and coolant layers were uniformly scaled.

As the overall thickness of the blanket increases, the TBR contribution from innermost layers (Breeder Layers 1 and 2) rise and layer 2 starts to stabilise. Deeper layers (Breeder Layers 3 and 4) contribute progressively less to the total TBR. The total TBR value quickly saturates at ≈ 450mm, and remains constant despite the breeding continuing to shift between layers.

To further characterise the layered blanket environment, cross-sectional heatmaps were generated to track neutron transport and its structural impacts. Note that the steel separators are included between each breeder-coolant interface. They are too thin to annotate, but note that gaps between the breeder and coolant indicate a separator.

Figure 6 maps the spatial distribution of the primary neutronic reactions driving the blanket's performance.

As seen in Figure 6a, tritium production is strictly confined to the designated breeder layers. The reaction density is highly asymmetric, with the vast majority of breeding occurring in the innermost layer immediately behind the first wall, before decaying sharply in the deeper radial zones.

Figure 6b demonstrates that neutron multiplication events follow a virtually identical spatial profile. The multiplication reactions are heavily concentrated at the front of the blanket, where the incident neutron flux first enters the system, and drop off rapidly in the subsequent outward layers.

Figure 7 displays the spatial distribution of both the total neutron flux and the resulting radiation damage.

As shown in Figure 7a, the total neutron flux attenuates deeply as it propagates radially through the blanket, becoming almost entirely exhausted by the rear boundary. Notably, this attenuation exhibits a distinct, step-like profile, characterised by sharp decreases in flux immediately across each discrete coolant layer.

Figure 7b maps the corresponding structural damage. As expected, the damage is heavily concentrated on the plasma-facing first wall. Beyond this initial boundary, the damage profile declines exponentially through the blanket depth, tracking the overall attenuation trend of the incident neutron flux.

Finally, the spatial distributions of neutron scattering and nuclear heating within the layered architecture were evaluated (Figure 8).

Figure 8a maps the frequency of neutron scattering events. The scattering density exhibits distinct, localised peaks that align directly with the positions of the internal coolant layers.

This spatial scattering pattern translates directly into the thermal energy deposition shown in Figure 8b. The nuclear heating profile follows a similar, albeit less pronounced, distribution peaking around the coolant layers. The highest overall concentration of heating is observed specifically within the second coolant layer.

5.2. Thickness Distribution Neutronics

To isolate the neutronic impact of spatial material distribution, the linear gradient parameter (ζ) was varied to shift the blanket from a forward-weighted to a rear-weighted geometry (see Figure 2).

As illustrated in Figure 9a, the global TBR remains relatively stable but exhibits a slight overall decrease as the blanket becomes rear-weighted ($\zeta > 0$). However spatial distribution of tritium production significantly shifts; inner plasma-facing layers dominate breeding in forward-weighted configurations ($\zeta < 0$), whereas deeper layers contribute a progressively larger fraction as ζ increases.

Figure 9b shows the neutron multiplication rate across the same domain. The total multiplication rate within the blanket decreases monotonically as the geometry shifts rearward. Conversely, the minor multiplication contribution from the structural separators slightly increases in back-loaded configurations.

In addition to tritium breeding, the structural impact of varying the linear gradient parameter (ζ) was assessed by tallying the neutron displacement damage (DPA) and the total damage energy across the blanket components.

As shown in Figure 10a, the global DPA decreases as the design becomes more forward weighted ($\zeta < 0$). As the material distribution is shifted rearward ($\zeta > 0$), the displacement damage experienced by the structural components steadily increases.

A similar trend is observed in the total damage energy (Figure 10b). The total damage energy deposited across the blanket is minimised in forward-weighted designs. Conversely, as the overall blanket damage energy decreases with negative ζ values, the specific energy deposited directly into the plasma-facing first wall increases.

The volume-integrated flux and resulting scattering events across the blanket components were evaluated as a function of the linear gradient parameter (ζ).

Figure 11a shows that the total volume-integrated flux within the blanket is maximised in forward-weighted configurations ($\zeta < 0$) and decreases as the geometry becomes rear-weighted ($\zeta > 0$).

The total scattering density (Figure 11b) exhibits a strongly correlated trend, decreasing as ζ becomes positive. Conversely, both the flux and scattering events localised within the structural layers increase in back-loaded configurations.

5.3. ML Optimisation Findings

The optimal blanket configurations identified by the Bayesian optimiser under various dimensional constraints are presented in Figure 12.

The fully unconstrained 8D configuration achieved the highest overall TBR of 1.4055 by pushing all parameters to their binding limits: maximising all breeder layers to 16.0 cm and minimising all coolant layers to 1.5 cm. The partially constrained 4D configurations exhibited lower overall breeding performance. Restricting the optimiser strictly to the maximum breeder budget (TBR: 1.2805) or the minimum coolant budget (TBR: 1.2080) resulted in consecutive decreases in the maximum achievable breeding ratio. Both were similarly under binding conditions, where the 4D coolant study pushed all the coolant thickness as to the outermost layers as much as allowed.

When restricted to the 44.0 cm total thickness budget, the ML model contrastingly identified an “Exponentially decaying” configuration where the breeder layer thicknesses decayed radially outward (16.0 cm, 10.5 cm, 6.9 cm, and 4.6 cm), maintaining minimised coolants, yielding a TBR of 1.3304. By comparison, the manually defined binding constraint configuration yielded an almost statistically identical TBR of 1.3303.

For clarity in evaluating the high-dimensional space, the discrete blanket layers are denoted sequentially from the plasma-facing first wall outwards: breeder layers are labelled B1 through B4, and their adjacent coolant layers are labelled C1 through C4.

To visualise the parameter landscape, 2D projections of the unconstrained 8D surrogate model were generated (Figure 13). These slices plot the thickness of the innermost breeder layer (B1)—which exhibited the strongest single-variable influence on the Tritium Breeding Ratio (TBR)—against other selected layers.

The gradients within these projections directly illustrate parameter sensitivity. Highly slanted contour lines, such as those mapping B1 against the secondary breeder (B2, Figure 13a) or the inner coolant (C1, Figure 13c), indicate that the TBR is sensitive to variations in both plotted variables. Here the predicted maximum TBR consistently isolates at the boundary corners, favouring maximised breeder thicknesses and minimised coolant thicknesses.

Conversely, projections mapping B1 against the outermost layers, such as B4 (Figure 13b) and C4 (Figure 13d), display nearly vertical contour lines. This indicates that the global TBR is highly insensitive to dimensional changes in the outermost blanket layers compared to the innermost layers.

6. DISCUSSION

Figure 5 demonstrates that tritium breeding is predominantly concentrated in the region closest to the "front" (facing the plasma), the inner blanket layers.

7. CONCLUSION

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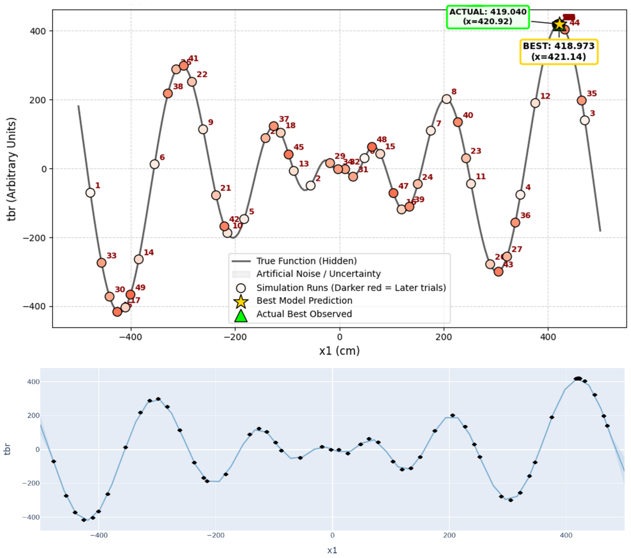
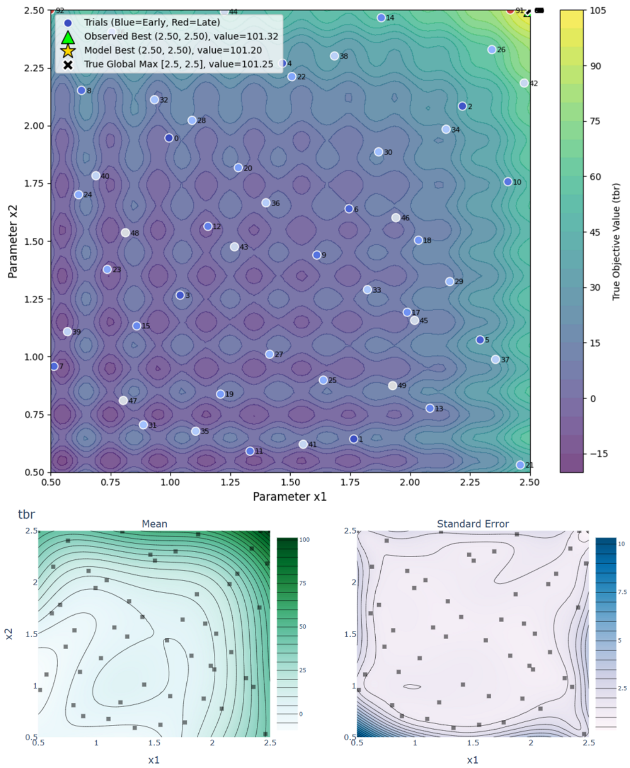
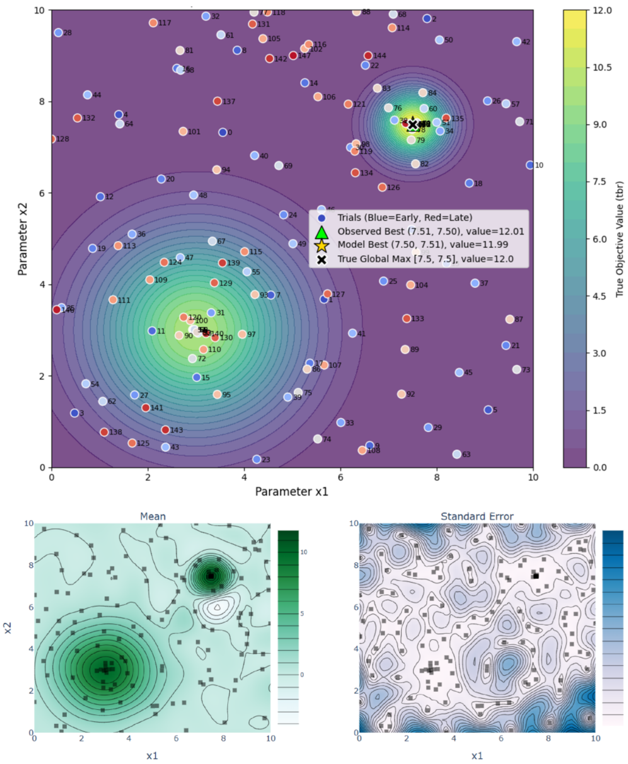


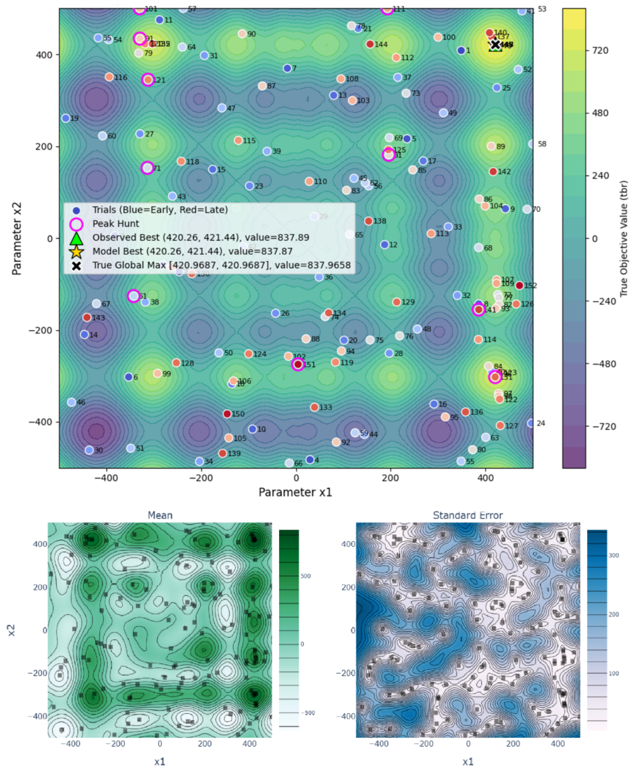
(a) 1D multi-modal Schwefel function test



(b) 2D Gramacy & Lee function test

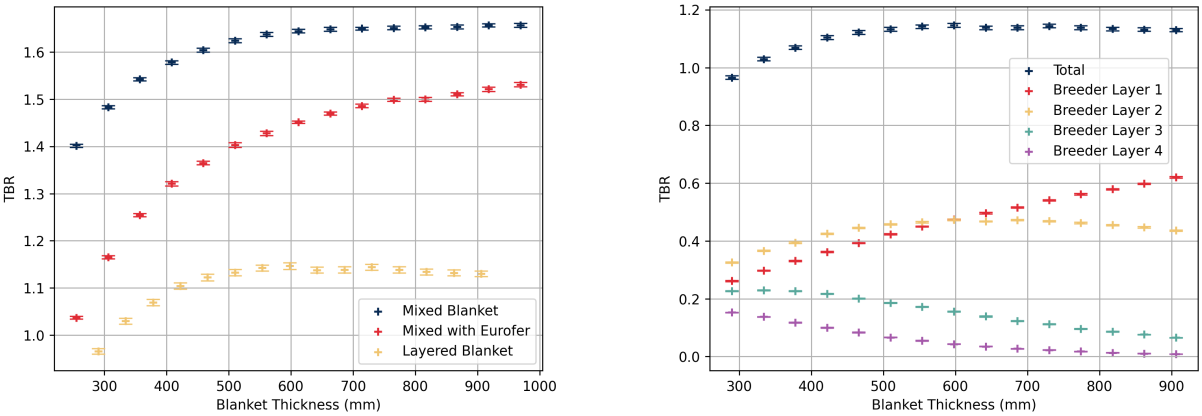


(c) 2D Hidden Peak function test



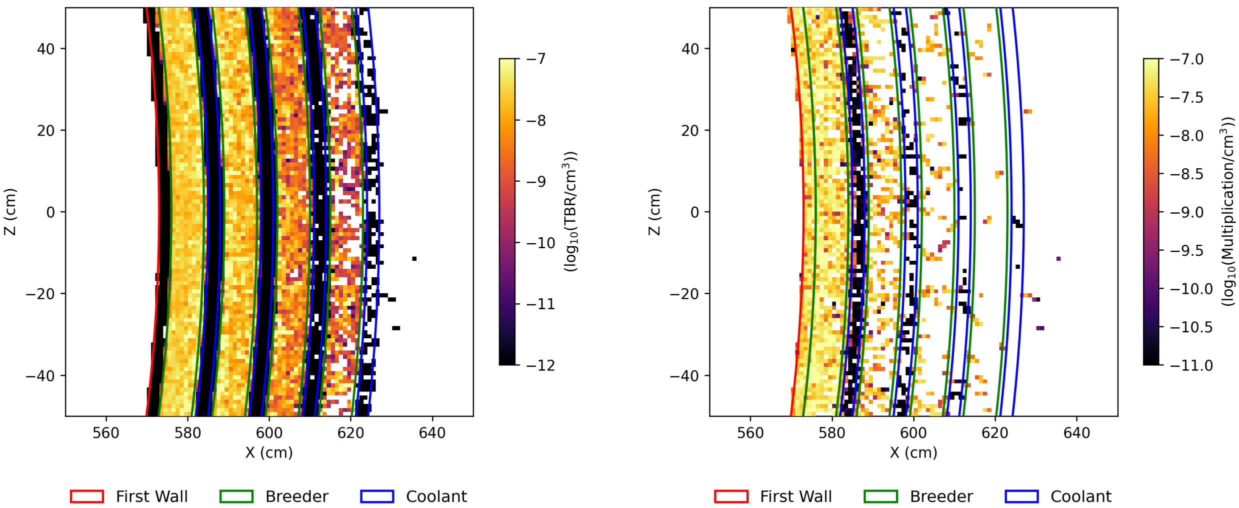
(d) 2D multi-modal Schwefel function test

Figure 4: Comprehensive benchmarking of the ML optimisation algorithm across multiple challenging landscapes. (a) A 1D Schwefel test case illustrating full surrogate model reconstruction against the true function. The uncertainty in the model is plotted but too small to be visible. (b), (c) and (d) demonstrate various 2D function tests of the optimiser, where the final mean surrogate model of the space and its paired standard deviation is plotted underneath each. (a) and (b) used $N = 100$ trials whereas (c) and (d) used $N = 150$, where this includes $N_{sobol} = 50$ sobol trials for all. UCB with $\beta = 1,000,000$ was consistent across all trials. Random injection and multi-modal exploitation was implemented for (c) and (d) with $\epsilon = 0.75$ and $n_{hunt} = 10$. (d) includes pink markers to annotate where a secondary "peak hunt" is triggered.



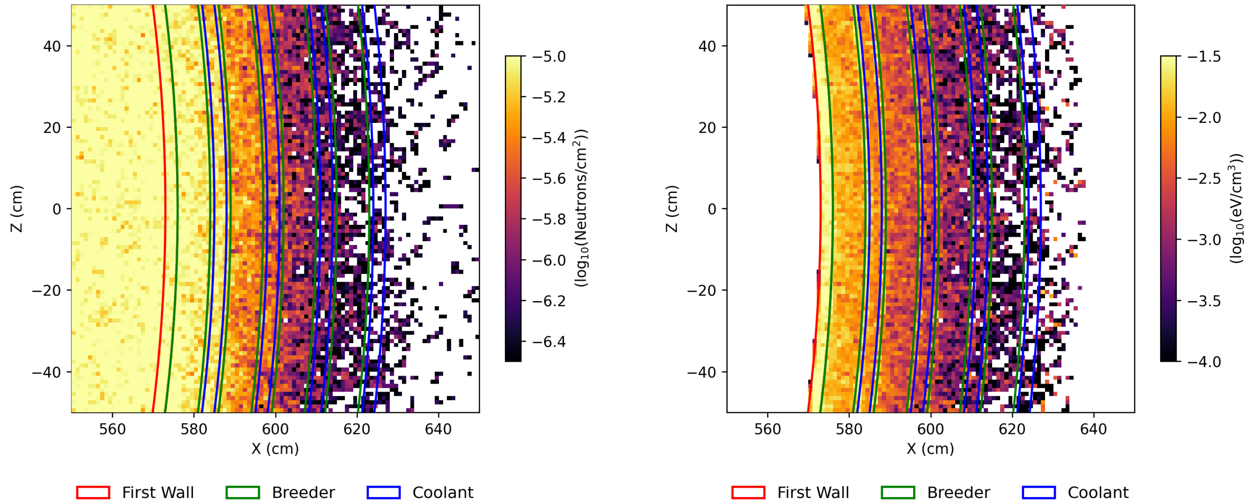
(a) TBR comparison across modelling approximations **(b)** Layer-wise TBR contribution vs. thickness

Figure 5: Evaluating TBR of the layered blanket architecture. (a) Highlights the degradation of global TBR when transitioning from idealised homogeneous mixtures to realistic heterogeneous layers. (b) Demonstrates the spatial distribution of tritium production, showing heavy concentration in the plasma-facing layers as overall blanket depth increases.



(a) Tritium production density **(b)** Neutron multiplication density

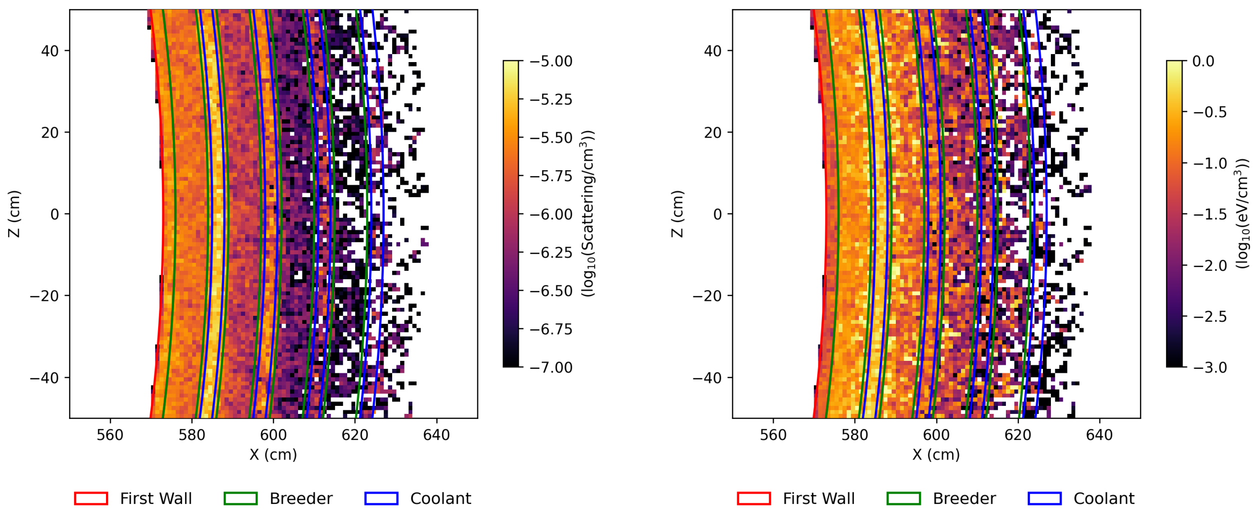
Figure 6: Cross-sectional heatmaps detailing the spatial distribution of key neutronic reactions. (a) Displays the density of tritium breeding reactions, strictly localised within the discrete breeder layers. (b) Illustrates the concentration of neutron multiplication events across the blanket depth.



(a) Total neutron flux distribution

(b) Radiation damage distribution

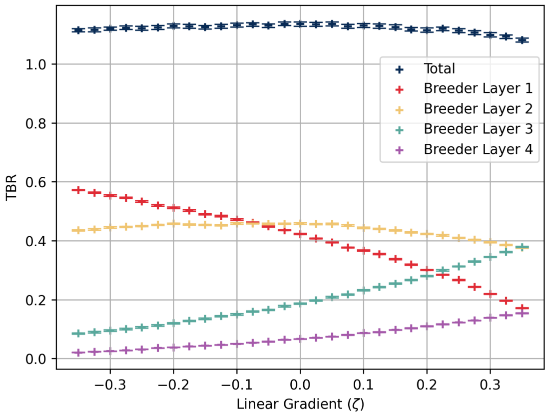
Figure 7: Cross-sectional heatmaps of flux and radiation damage. (a) Illustrates the progressive attenuation of the neutron flux across the material boundaries. (b) Shows the spatial distribution of structural radiation damage.



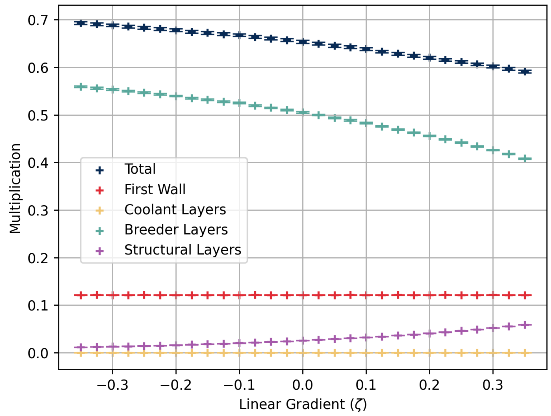
(a) Neutron scattering density

(b) Nuclear heating distribution

Figure 8: Cross-sectional heatmaps of scattering and energy deposition. (a) Highlights distinct peaks in neutron scattering density localised around the discrete coolant layers. (b) Displays the corresponding nuclear heating profile, noting a specific concentration around the second coolant layer.

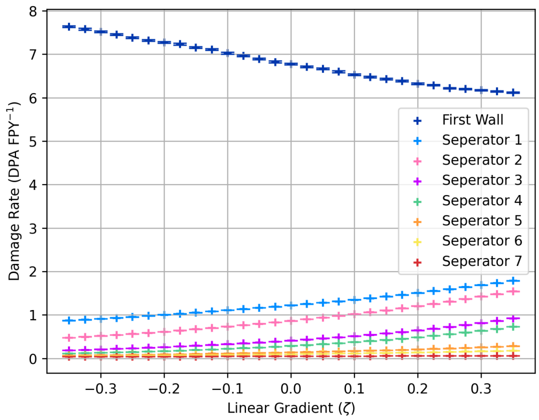


(a) TBR layer contribution

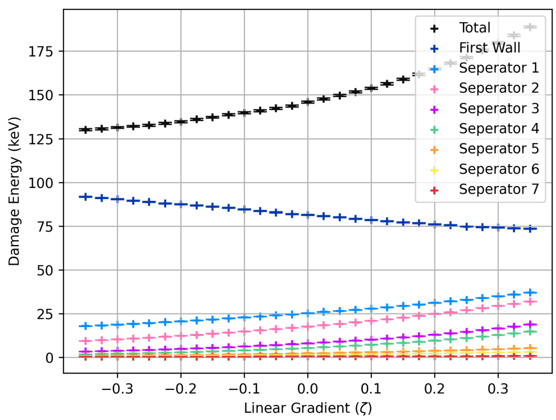


(b) Neutron multiplication contribution

Figure 9: The impact of the linear gradient parameter (ζ) on key neutronic performance metrics. (a) Tracks the total TBR and the individual contributions from discrete breeder layers. (b) Illustrates the total neutron multiplication alongside the specific contributions from the first wall, breeder, coolant, and structural separator layers. Statistical uncertainties are plotted, though error bars are generally smaller than the data markers.

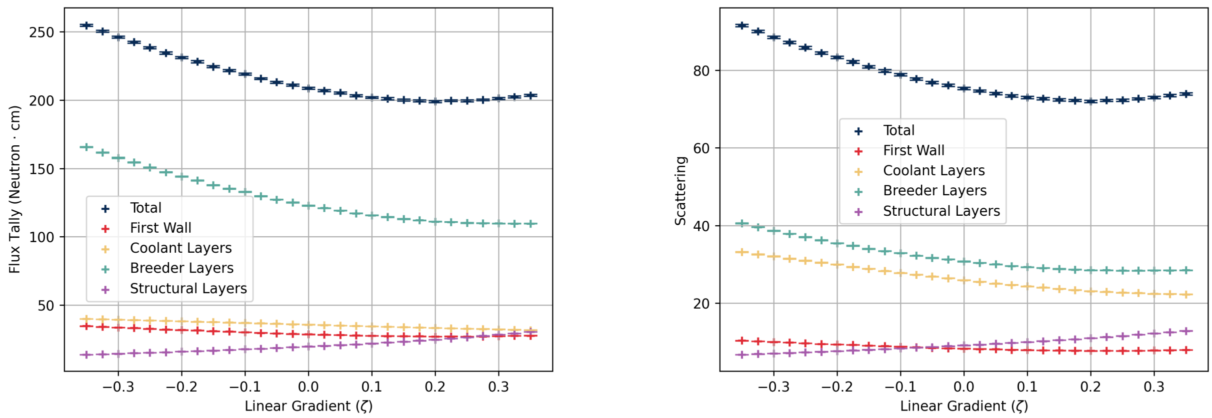


(a) DPA rates across components



(b) Damage energy deposition

Figure 10: The impact of the linear gradient parameter (ζ) on structural degradation metrics. Figure 10a tracks the Displacements Per Atom (DPA), while Figure 10b tracks the damage energy deposited into the first wall and internal structural separators. Statistical uncertainties are plotted, though error bars are generally smaller than the data markers.



(a) Volume-integrated flux tally

(b) Neutron scattering events

Figure 11: The effect of the linear gradient parameter (ζ) on neutron transport and interaction metrics. Figure 11a plots the volume-integrated flux (total neutron path length in n · cm) across the blanket and its individual components. Figure 11b plots the corresponding number of neutron scattering events. Statistical uncertainties are plotted, though error bars are generally smaller than the data markers.

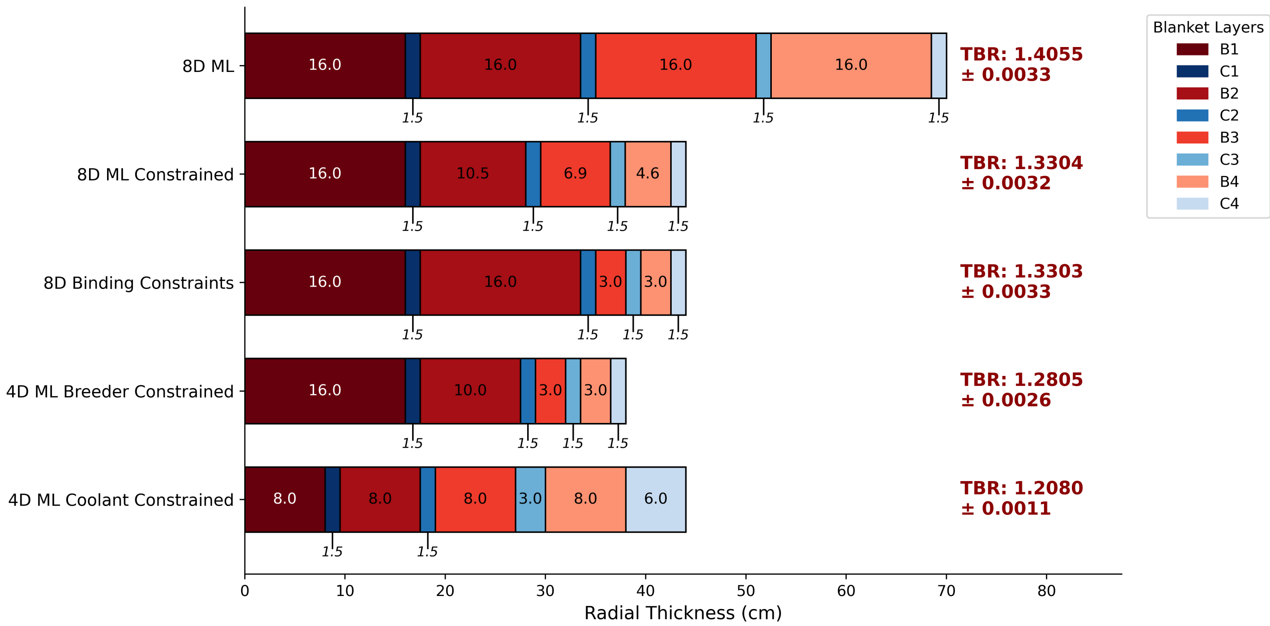
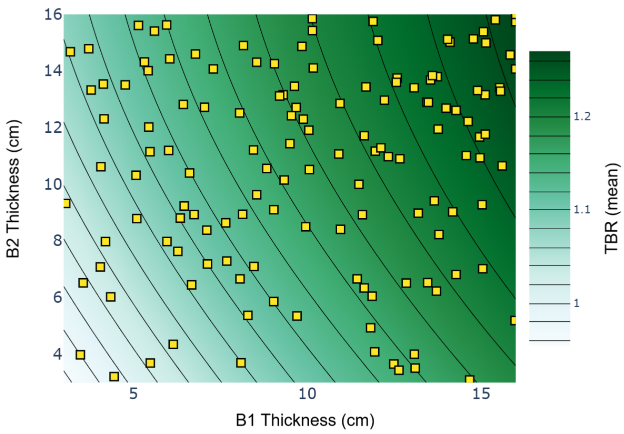
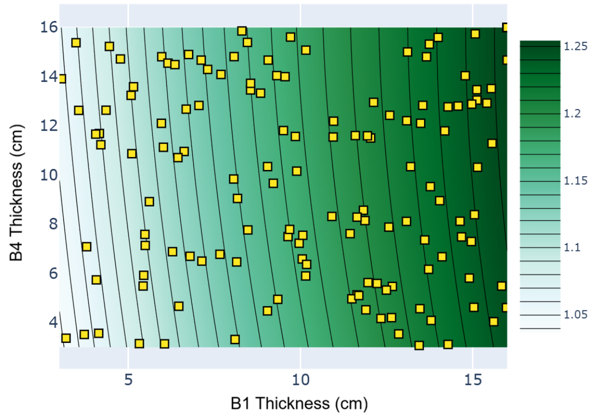


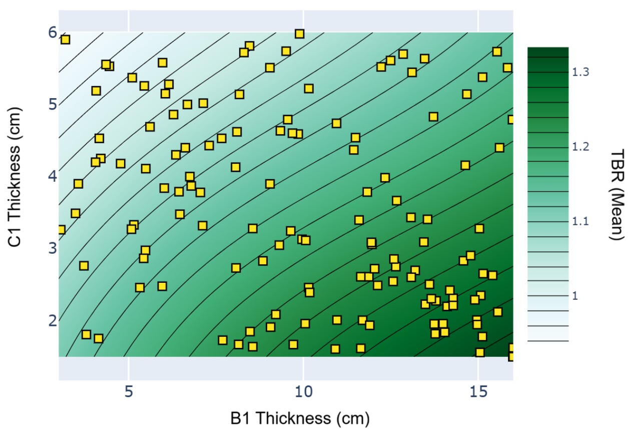
Figure 12: Radial thickness profiles and resulting global TBR for the ML optimised TBR-thickness studies detailed in Section 4.3.3. Associated TBR values are annotated alongside their standard statistical uncertainty. $N = 100$ was used for all except for unconstrained 8D $N = 150$, where these include $N_{sobol} = 50$. Hyper-parameters $\epsilon = 0.5$ and $\beta = 1,000,000$ were tuned for these studies. The total runtime for every optimisation was between $1 - 1.5$hrs, limited by the neutronics code taking 1 minute per iteration, not the ML optimiser.



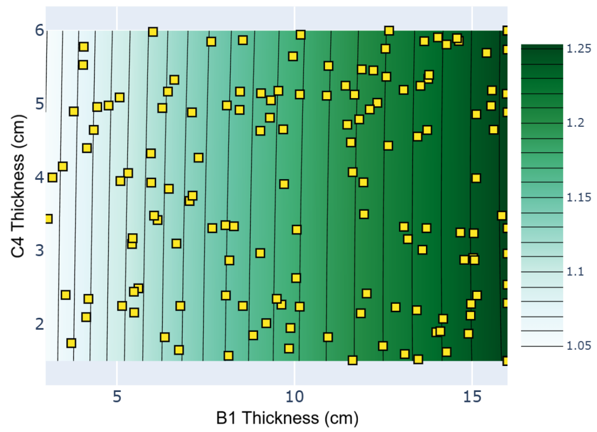
(a) B1 vs. B2 thickness



(b) B1 vs. B4 thickness



(c) B1 vs. C1 thickness



(d) B1 vs. C4 thickness

Figure 13: 2D contour projections of the 8D surrogate model from the unconstrained optimisation run. Each panel plots the thickness of the innermost breeder layer (B1) against a secondary layer (B2, B4, C1, or C4), with all non-plotted dimensions held constant at their optimal boundary limits.